

Hierarchical Core Execution Model • Canonical DAG & Role Responsibilities

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|   `-- agent/researcher/skill/SKILL.md
|   |
|   |   |-- source authority needed
|   |   |   `-- source-policy.md
|   |
|   |   |-- evidence/provenance needed
|   |   |   `-- evidence-guidance.md
|   |
|   |   |-- fixture/success derivation needed
|   |   |   `-- fixture-guidance.md
|   |
|   |   `-- example/reference needed
|   |       `-- examples.md
|
|-- WHEN repository work is required
|   |
|   v
|   resolve required Tool / Capability
|   |
|   |   |-- repository-read
|   |   |-- Scan
|   |   `-- Discovery when required
|   |
|-- WHEN external research is required
|   |
|   v
|   resolve required Skill / Tool / Provider
|   |
|   |   |-- LOAD only required resource
|   |   |-- execute research
|   |   |-- collect evidence
|   |   `-- preserve provenance
|   |
|-- WHEN parallel research is required
|   |
|   v
|   LOAD parallel capability
|   |
|   |   |-- execute bounded parallel research
|   |   |-- collect results
|   |   `-- consolidate results
|   |
|-- evidence processing
|   |   |-- agent/researcher/evidence/consolidate.ts
|   |   `-- agent/researcher/evidence/provenance.ts
|   |
|-- synthesis
|   |   `-- agent/researcher/synthesis/research-synthesis.ts
|
|-- OUTPUT
|   |
|   v
Researcher(id)
|
|-- plan_id
|   |
|   |   |-- Design
|   |   |   |-- required design
|   |   |   |-- affected responsibility
|   |   |   |-- affected component
|   |   |   |-- target when established
|   |   |   |-- dependency
|   |   |   |-- constraint
|   |   |   `-- expected result
|   |
|   |   |-- TODO / Action
|   |   |   |
|   |   |   |-- task_1
|   |   |   |   |-- action
|   |   |   |   |-- target

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| | | |-- dependency
| | | |-- constraint
| | | |-- expected result
| | | |
| | | |-- only when required
| | | |   |-- Skill
| | | |   |-- Tool
| | | |   |-- Capability
| | | |   |-- file
| | | |   |-- script
| | | |   |-- schema
| | | |
| | | |-- task_2
| | | |-- task_n
| | | |
| | |-- Requirements
| | | |-- requirement_1
| | | | |-- statement
| | | | |-- evidence
| | | | |-- target
| | | | |-- dependency
| | | | |-- constraint
| | | | |-- expected result
| | | |
| | |-- requirement_n
| | |
|-- schema_id
| | |-- required structure
| | |-- required contracts
| | |-- required properties
| | |-- constraints
| | |-- schema/reference when required
| |
|-- fixture_id
| | |-- validation case
| | |-- related success criterion
| | |-- input/reference
| | |-- assertion/condition
| | |-- expected result
| |
|-- goal_id
| | |-- required outcome
| | |-- why it matters
| | |-- success criteria
| | |-- supporting evidence
| |
|-- validation_id
| | |-- what must be validated
| | |-- required input
| | |-- required condition
| | |-- required proof
| |
|-- <additional_name>_id ... n
| | |-- expand recursively only when required
| | | |-- requirement
| | | |-- action
| | | |-- dependency
| | | |-- constraint
| | | |-- resource
| | | |-- expected result
| |
|
v
plan_id
|
v
PLANNER
|
|-- ROLE
| | |-- agent/planner/role.ts
| | |-- agent/planner/workflow.ts

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|
|-- INPUT
|   |-- plan_id
|
|-- LOAD SKILL when Planner work requires it
|   |-- agent/planner/skill/SKILL.md
|
|-- REVIEW
|   |-- LOAD review-guidance.md when required
|   |-- evidence sufficiency
|   |-- file coverage
|   |-- structure
|   |-- design
|   |-- goal clarity
|   |-- fixture usability
|   |-- unresolved conflicts
|
|-- implementation support
|   |-- agent/planner/review/survey.ts
|   |-- agent/planner/review/review-report.ts
|   |-- agent/planner/verification/research.ts
|   |-- agent/planner/verification/structure.ts
|
|-- PLAN WORK
|   |-- LOAD plan-guidance.md when required
|   |-- inspect plan_id
|   |-- identify required refinement
|   |-- preserve every required branch
|
|-- OUTPUT
|   |
|   v
|   audit_id
|       |-- findings
|       |-- design issues
|       |-- structure issues
|       |-- requirement gaps
|       |-- required refinement
|
|
v
REFACTOR / REFINEMENT
|
|-- ROLE
|   |-- agent/refactor/role.ts
|   |-- agent/refactor/workflow.ts
|
|-- INPUT
|   |-- plan_id
|   |-- audit_id
|
|-- LOAD SKILL when required
|   |-- agent/refactor/skill/SKILL.md
|
|-- LOAD transformation-guidance.md
|   |-- when transformation is required
|
|-- LOAD scope-guidance.md
|   |-- when scope confirmation is required
|
|-- refine SAME plan_id
|-- preserve provenance
|-- preserve required branches
|
|-- OUTPUT
|   |
|   v
|   updated plan_id
|
v

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GAP ANALYSIS
|
|-- inspect updated plan_id
|
|-- gap_1 ... gap_n
|   |-- affected branch
|   |-- missing requirement
|   |-- required correction
|   |-- expected resolved state
|
|-- fix gaps
|-- recheck
|
|-- gap_0
|   |
|   v
FINAL plan_id
|
|-- original requirements preserved
|-- required refinement complete
|-- dependencies satisfied
|-- constraints preserved
|-- expected results defined
|
|-- INVARIANT
|   |-- FINAL plan_id >= Job requirements
|
|
v
EVALUATION
|
|-- when applicable case exists
|   |-- evaluate FINAL plan_id
|   |-- compare expected behavior
|   |-- preserve result
|
|
v
TRIPLE VALIDATION
|
|-- validation_id
|
|-- SCHEMA VALIDATION
|   |-- schema_id
|   |-- FINAL plan_id
|   |-- schema/plan/artifact/plan.schema.json
|   |-- validation/schema/validator/schema_validator.py
|   |-- schema_validation(hash)
|
|-- FIXTURE VALIDATION
|   |-- fixture_id
|   |-- FINAL plan_id
|   |-- validation/fixture/validator/fixture_validator.py
|   |-- fixture_validation(hash)
|
|-- GOAL VALIDATION
|   |-- goal_id
|   |-- FINAL plan_id
|   |-- validation/goal/validator/goal_validator.py
|   |-- goal_validation(hash)
|
|
v
Final_Confirmed_Validation(hash)
|
|-- FINAL plan_id CONFIRMED
|
|
v
PROMOTE
|
|
v
Researcher(id)
|

```

```
|-- plan_id      [OK] CONFIRMED
|-- schema_id   [OK] CONFIRMED
|-- fixture_id  [OK] CONFIRMED
|-- goal_id     [OK] CONFIRMED
|-- validation_id [OK] CONFIRMED
`-- <required>_id [OK] CONFIRMED
|
v
Researcher(id) FINAL
|
| update JOB_ID record
|
|-----> JOB_ID
|         |-- Prompt_id [OK]
|         `-- Plan_id   [OK] CONFIRMED
|
v
Builder
|
Prompt_id + Plan_id
v
Builder Execution
v
Build Result
v
Hash Verification
v
DONE
```